

Leveraging LDP for High-Trust OpenSearch UBI

Tuning search when you can't store the queries

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OpenSearchCon NA
September 24, 2026

About me

- Independent consultant, Mountain Fog
- OpenSearch UBI and opensearch-migrations maintainer
- Apache Software Foundation member, OpenNLP PMC Chair
- PII redaction tooling at Philterd
- Works in search, NLP, and privacy

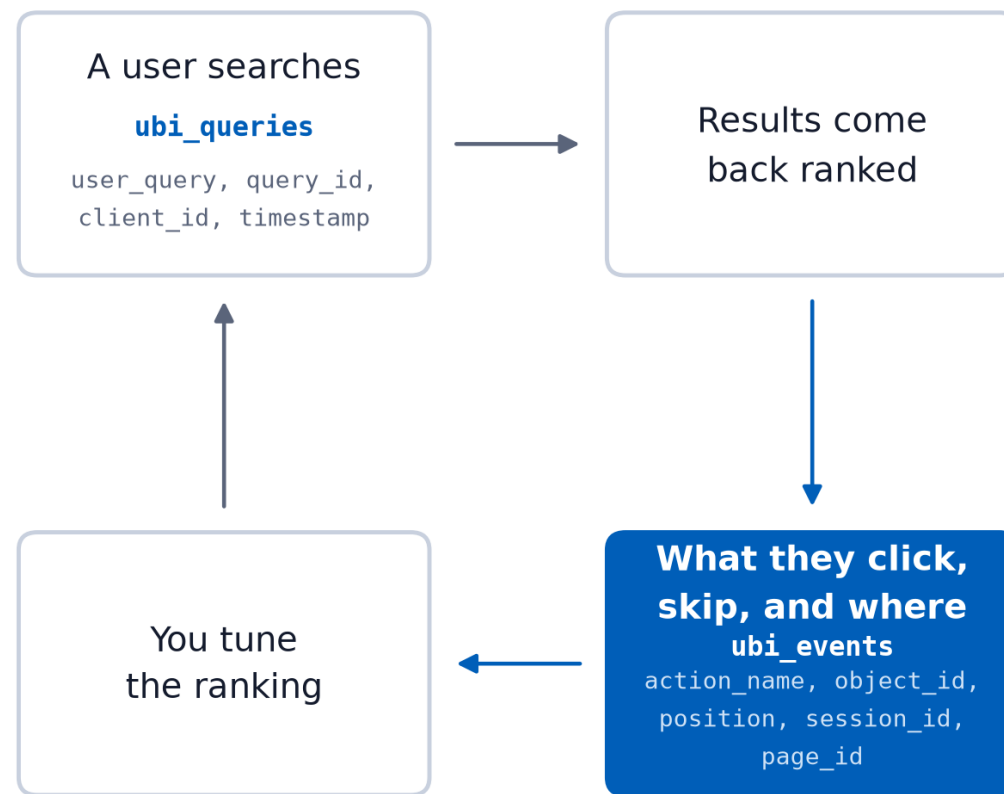
jeffzemerick.dev

PART 1

The problem

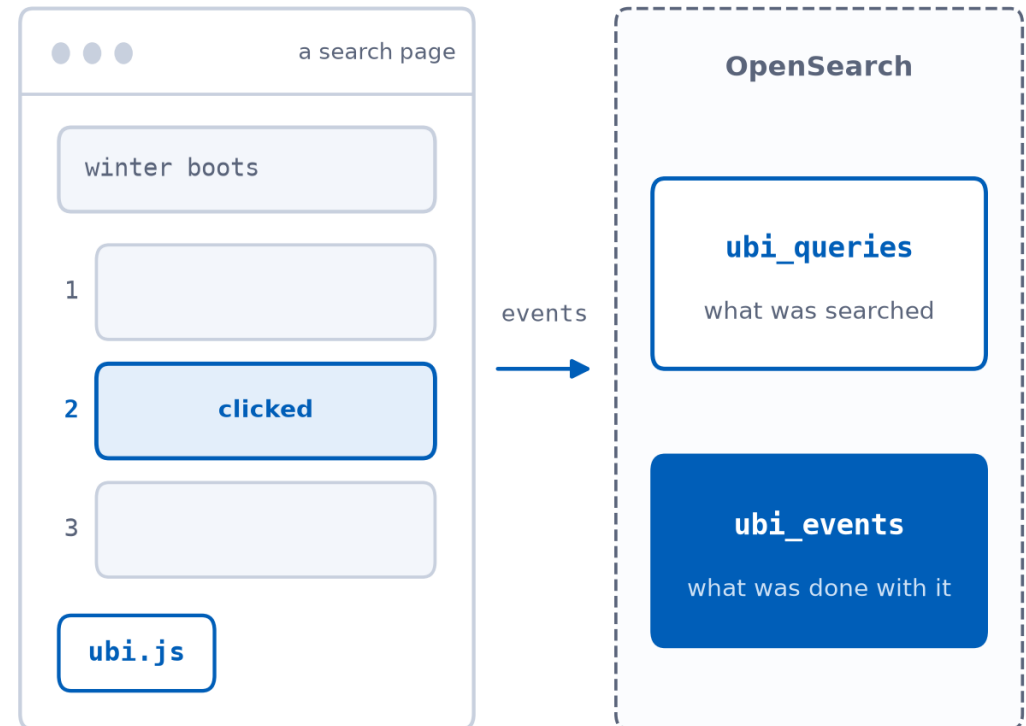
To tune search you have to see what users do

- Relevance is measured
- Which queries fire, which results get clicked, at which position
- Learning-to-rank trains on exactly this signal
- Without it you are guessing, and shipping guesses to production



OpenSearch UBI exists for this

- `ubi_queries`: `user_query`, `query_id`, `client_id`, `timestamp`
- `ubi_events`: `action_name`, `object_id`, `position`, `session_id`, `page_id`
- `ubi.js` collects interactions in the browser and ships them to the cluster
- This is the raw material relevance tuning runs on



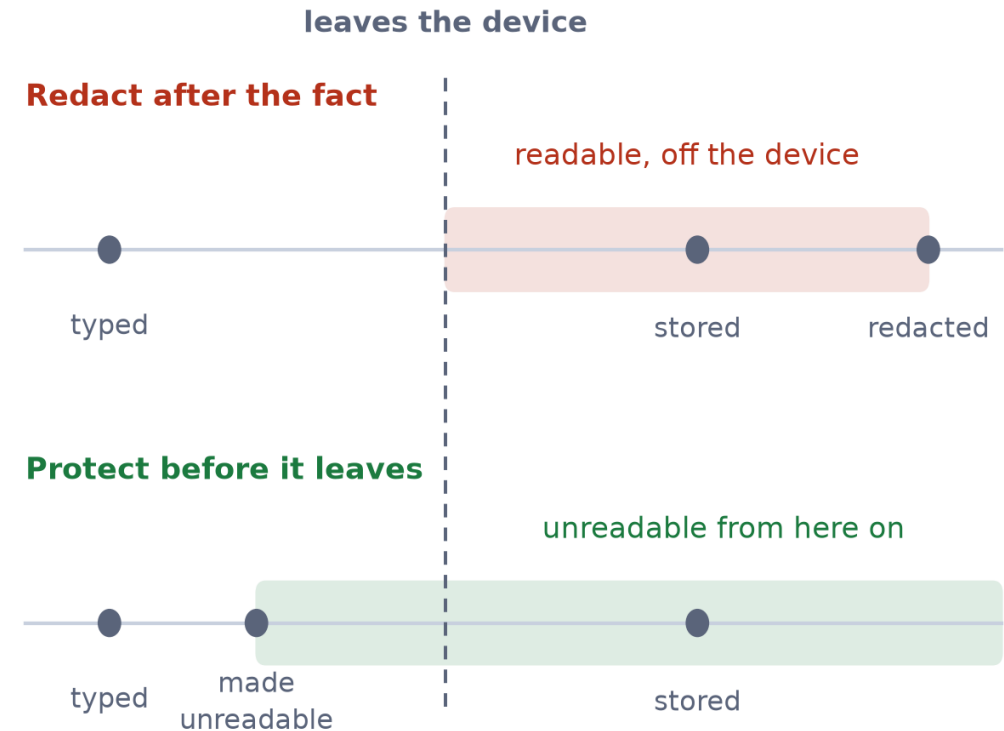
For examples and code, go to ubisearch.dev

Legal department sees the schema

- `user_query` is free text a person typed. It can contain anything
- `client_id` and `session_id` make it linkable across time
- In healthcare and finance, that combination is a hard stop
- Zero-trust stopped being a buzzword and became a blocker to relevance tuning

Redaction is reactive by construction

- You collect the PII first, then try to remove it
- You discover the failure after the fact, if at all
- A free-text query has no schema to redact against
- The fix has to be that the sensitive value never leaves the device readable



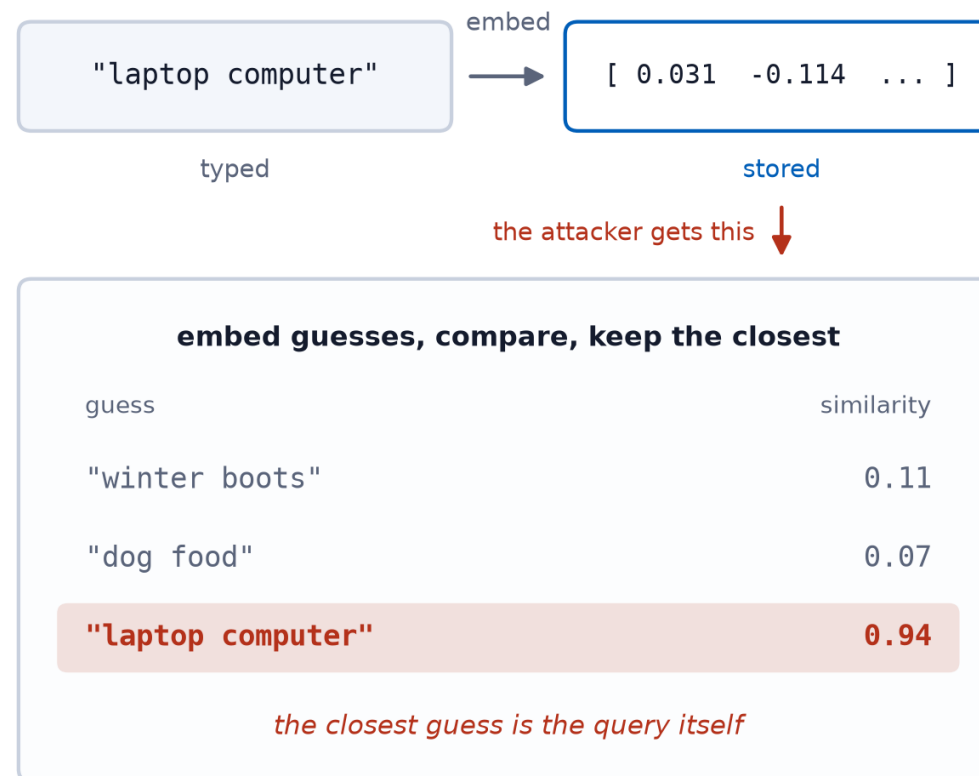
Still need to redact data at rest? philterd.ai

PART 2

Embeddings are not anonymization

The obvious first idea

- "We will not store the text. We will store the embedding."
- It feels private. It is a vector of floats, not words
- It is not private. It is a lossy encoding
- Vector inversion recovers the intent



LIVE NOTEBOOK

Demo: invert a raw query vector

Notebook sections 1 to 6, then section 9

The raw vector gives up the intent

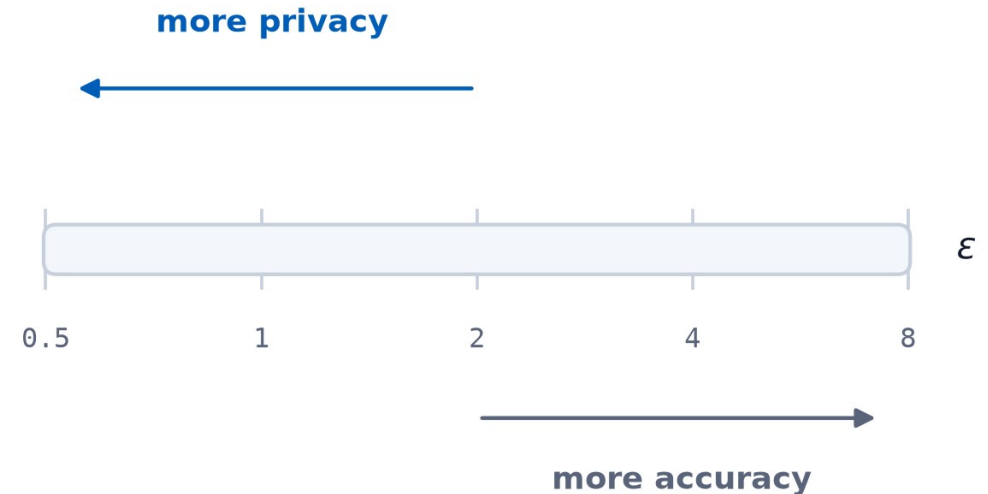
- Query: "laptop computer"
- Attacker inverts the raw vector, top hit: Laptop
- No text was stored, and the intent leaked anyway
- **Storing embeddings is not a privacy control**

PART 3

The fix and the proof

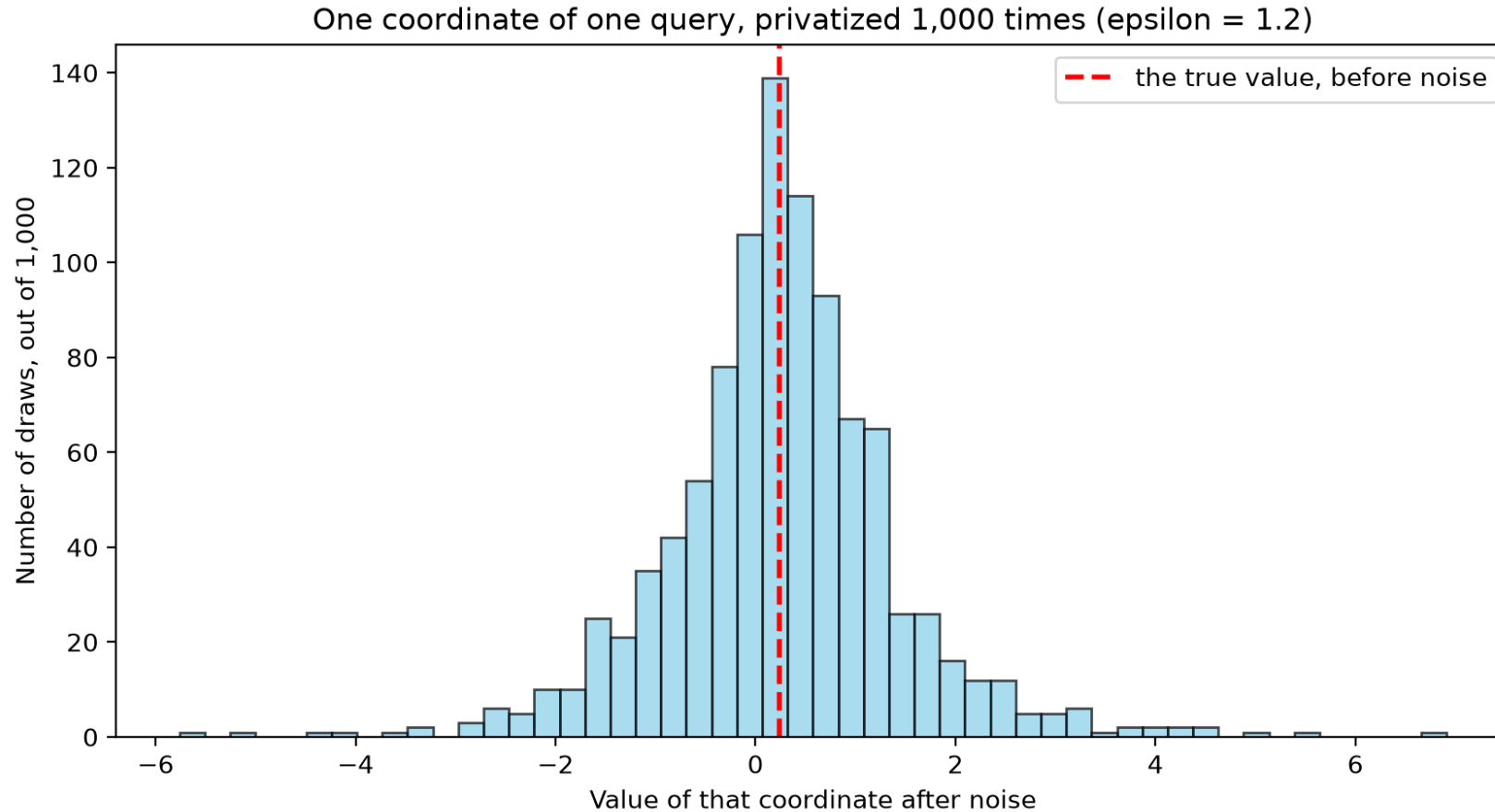
Local Differential Privacy

- The vector was never the problem, storing it readable was
- Add calibrated noise to the query vector on the device
- Epsilon is the dial, and lower epsilon means more noise and more privacy
- The cluster never receives a readable query, so there is nothing to scrub



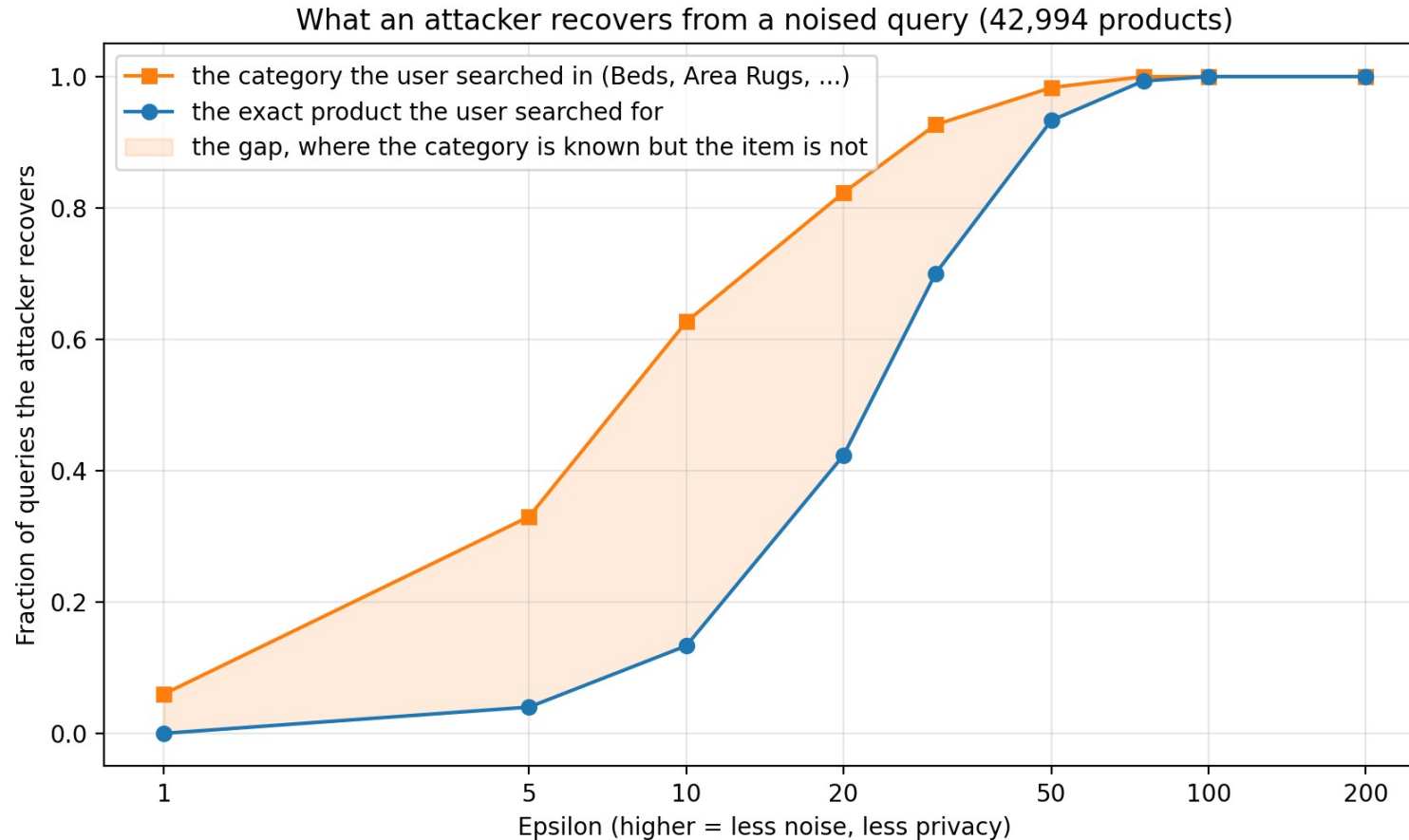
More on local differential privacy: en.wikipedia.org/wiki/Local_differential_privacy

Verification 1: the noise is what we claim



A biased mechanism would put the peak off the red line. The width is the privacy.

Verification 2: measure an actual attacker



The category leaks well before the item does. At epsilon 1, neither is usable.

Data: WANDS, Wayfair product search relevance (ECIR 2022), MIT licensed: github.com/wayfair/WANDS

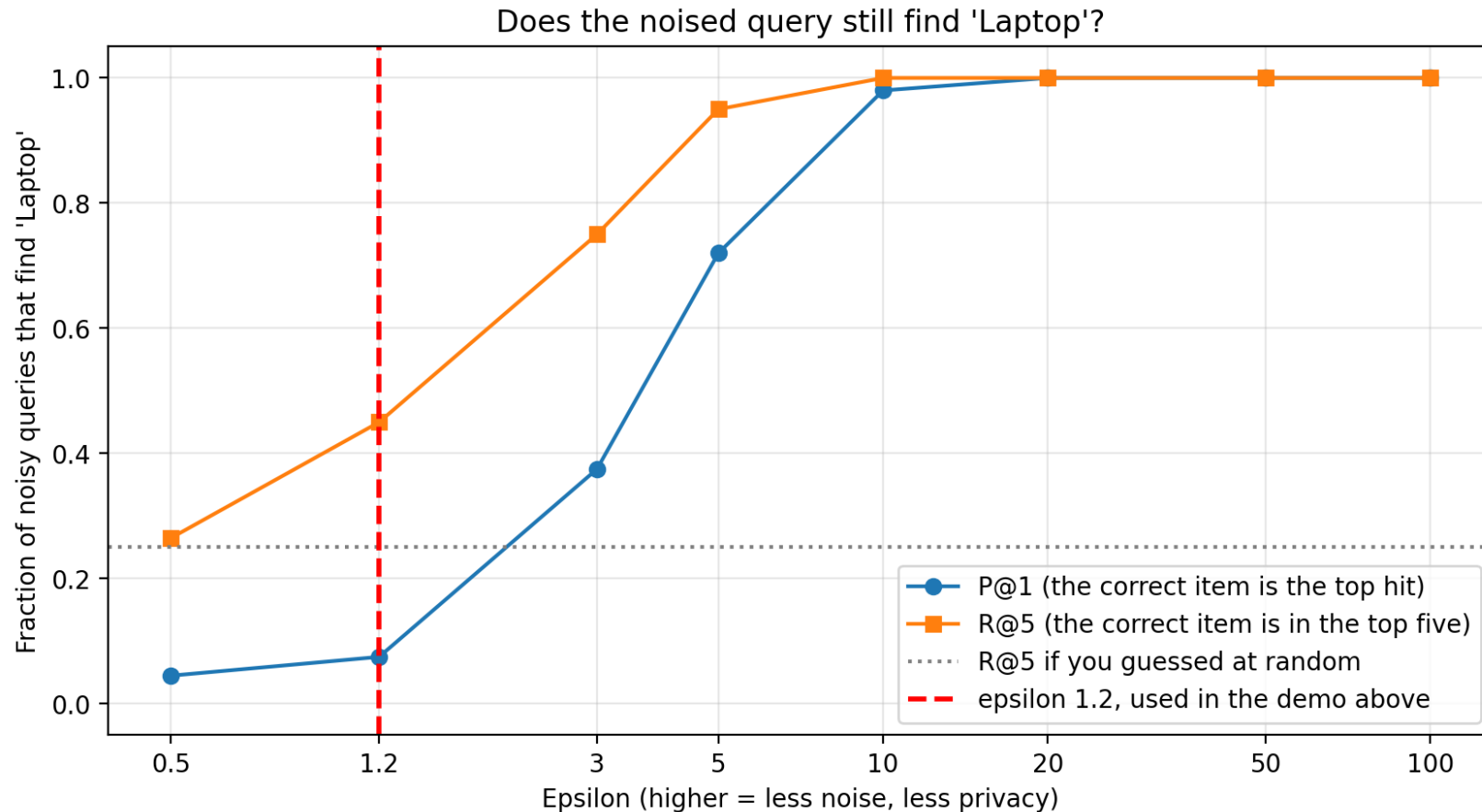
PART 4

What it costs

That first demo was rigged

- It ran at epsilon 1.2 on a 20 document index
- The attacker learned nothing, and neither did the user

Epsilon 1.2 is barely better than guessing



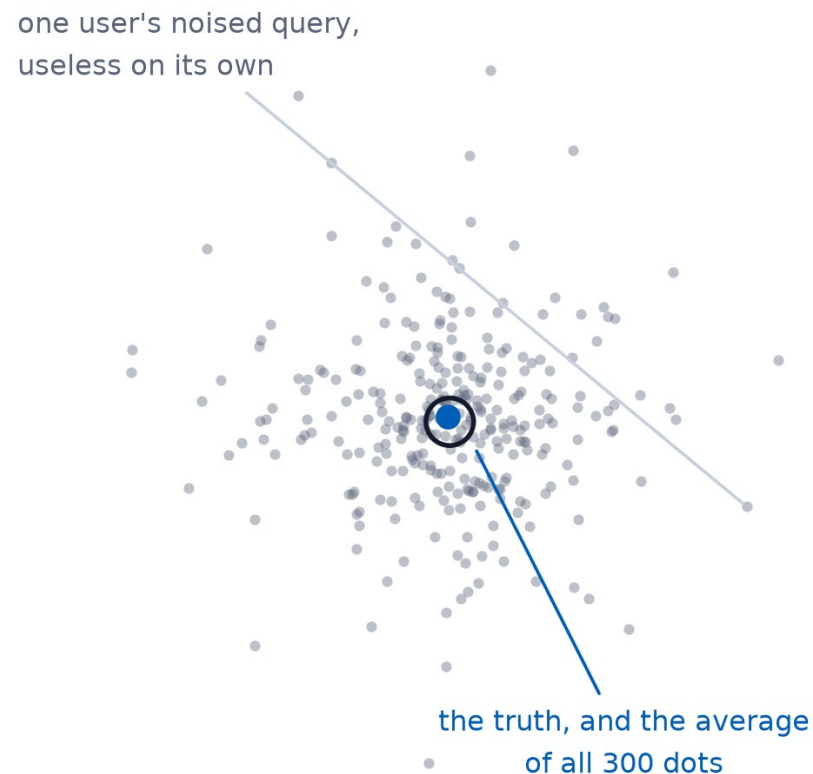
The red line is where the first demo ran. Utility only comes back where the attacker wins too.

PART 5

What you keep

The asymmetry is the whole point

- Laplace noise is zero-mean
- It cancels when you average over many independent users
- One person's query is unrecoverable
- The trend across ten thousand people is not
- That trend is all UBI needs to tune relevance



LIVE NOTEBOOK

Demo: segments of real users

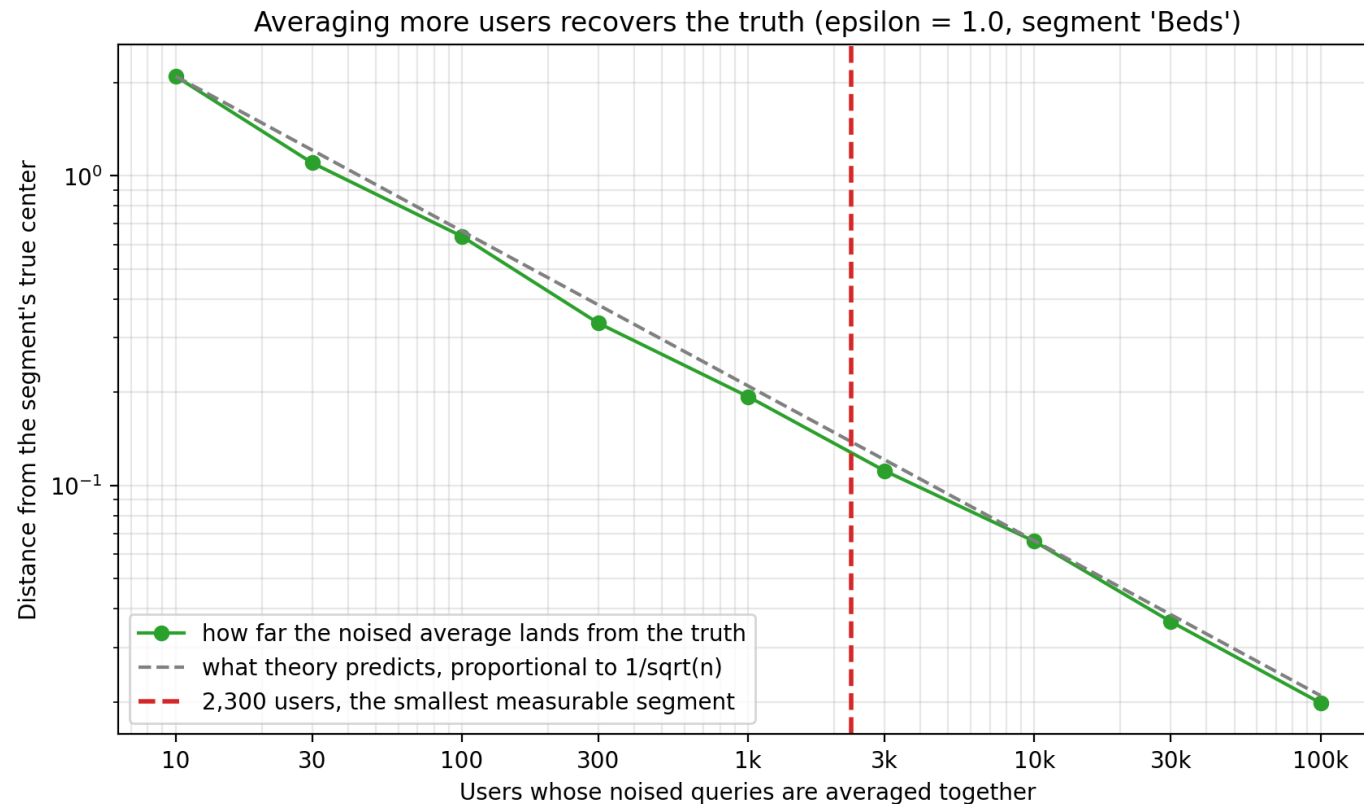
Notebook section 11

480 real queries from Wayfair's WANDS dataset, grouped into 5 segments

Every user privatizes independently, on their own device

Error falls as $1 / \sqrt{n}$

Average the noised queries of n users in one segment, then see how far that average lands from the truth.



The red line is the smallest segment you can measure.

How big a segment has to be

- Error drops below the spread of the index at roughly 2,300 users
- That is at epsilon 1, and halving epsilon needs four times the users
- Segments with thousands of users are measurable under LDP
- Segments with dozens are not measurable
- Your head terms work, but your long tail does not

LDP does not cost you your analytics.

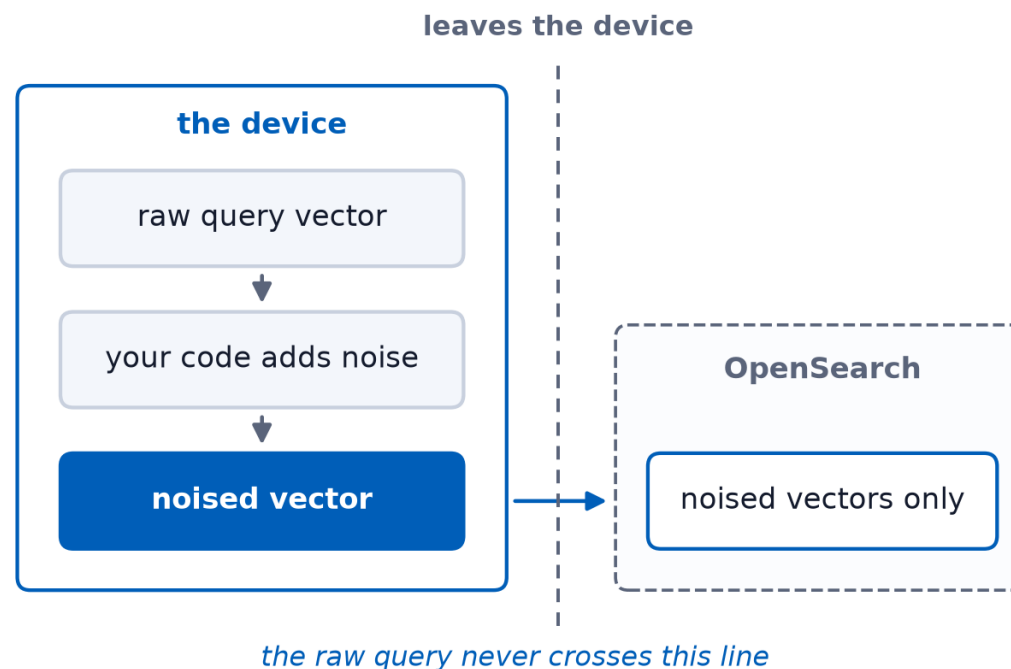
It costs you the ability to ask about one person.

PART 6

Back in OpenSearch

Where the noise gets injected

- On the device, before `ubi.js` sends the event
- Aggregation happens at query time over the noised data
- Nothing downstream needs to be trusted, because nothing downstream has the original
- You set epsilon in the client you ship



An upcoming UBI RFC will bring this capability into `ubi.js`

What changes in ubi_queries

today

```
{
  "query_id": "q-8f31c4",
  "client_id": "c-2f91",
  "timestamp": "2026-09-24T10:12:03Z",
  "user_query": "solid wood platform bed"
}
```



with LDP

```
{
  "query_id": "q-8f31c4",
  "client_id": "c-2f91",
  "timestamp": "2026-09-24T10:12:03Z",
  "user_query": "",
  "query_attributes": {
    "noised_query_vector": [0.03, -1.82, ...],
    "epsilon": 1.0
  }
}
```

UBI supports this today through query_attributes, and the RFC will propose making it first-class

What your dashboards can still compute

- Aggregate query intent per segment
- Demand trends and shifts over time
- Segment-level training signal for learning-to-rank
- What they cannot do is show you one user's session

The limitations

- The category leaks and that is worse for medicine than furniture
- What a given epsilon buys you depends on coordinate spread so it does not transfer between indexes
- Low-traffic segments stay unmeasurable
- This protects `user_query` but `ubi_events` still holds clicks and ids in the clear

Takeaways

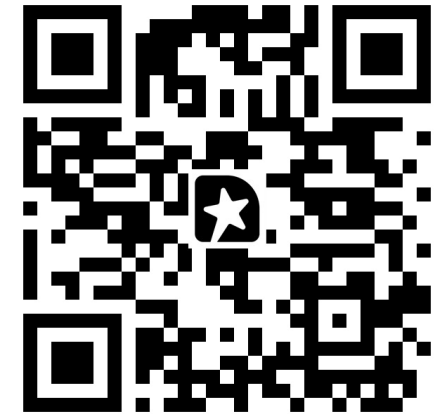
- Storing embeddings instead of text is not privacy
- Keeping sensitive data out of the index beats redacting it later
- Verify the mechanism by attacking it, not by trusting it
- You trade individual resolution for population accuracy, and relevance only needs the latter

Questions

github.com/jzonthemtn/ldp-for-search

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Session feedback